20. React Routing With Data API (Extra Topic - Not Mandatory)

▼ React Routing With Data API

- The **React Router Data API** was introduced in `React Router v6.4`.
- It allows us to:
  - Fetching Data before the page/UI renders.
- This makes routing **faster, cleaner, and more powerful**.

👉 Earlier:

- First Page Rendered ❌
- Then Data Fetched ❌
- Loader added ❌

👉 Now with Data API:

- Data loads first✅
- Then page renders with data✅
- No blank UI / No flash loading✅

▼ Key Features of the Data API:

1. **Route Definitions:**
   - Routes are defined as objects within an array, allowing for more flexible and modular configurations.

   > ***Note:*** `createBrowserRouter` ***is included in the Data API.***

   > ***Note:*** `BrowserRouter` ***is not included in the Data API.***

2. **Data Loading:**
   - Each route can have its own **loader function**

3. **Eliminating Render + Fetch Chains:**
    - By separating data fetching from rendering, the API helps to avoid common performance issues like render + fetch chains and the associated loading spinners.
4. **Flexibility:**
    - The Data API makes routing more flexible and adaptable, allowing for complex routing scenarios to be managed more effectively.
5. **New Router Types:**
    - The API introduces new router types, such as `createBrowserRouter`, which are designed to work with the data API.

▼ What is a `loader`?

- A **loader is just a function that fetches data before a page loads.**
- *Imagine you are browsing a website. When you browse to a new page, a loader or spinner is displayed to show that the data is being fetched until the new page loads.*
- Loaders in React Router work similarly.
- Loader runs **before** the component renders.
- Loader is attached to a **specific route.**
- **Loader Function Syntax:**

```
loader: async (props) => {
  console.log(props); // request, params, context
}
```

- **Example: Loader With API Call**

```
{
  path: "/users",
  element: <Users />,
  loader: async () => {
    let response = await fetch("https://api.github.com/users");
    let data = await response.json();
```

```
    return data;
  },
}
```

This loader:

- Fetches GitHub users
- Sends the data to the Users component

▼ What is `useLoaderData` Hook?

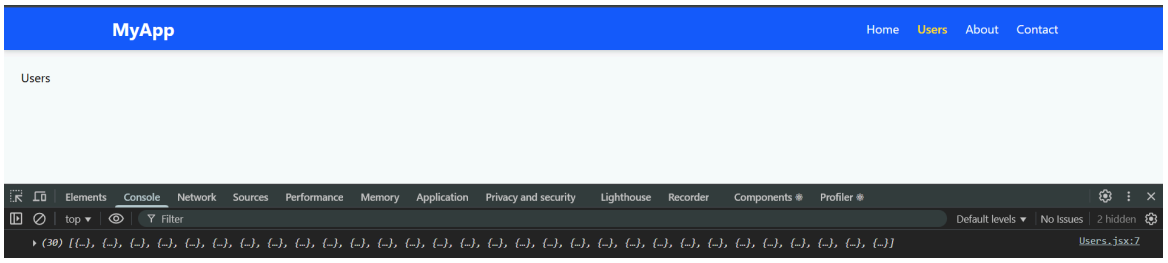
- The `useLoaderData` is a React hook provided by React Router DOM library.
- The `useLoaderData` is used to **get the data returned by the loader** inside the component.
- It avoids writing:
  - ❌ useEffect
  - ❌ extra states
  - ❌ manual loading logic

▼ Accessing the data provided by the `loader function` by using `useLoaderHook`

- The data provided by the loader function inside the Users component route configuration we now have to access that data by using the hook called as `useLoaderData` which is provided by the ***react-router-dom*** library.
- **Code: Users.jsx**

```
import React from "react";
import { useLoaderData } from "react-router-dom";

const Users = () => {
  //fetching data by using the useLoaderData Hook
  let data = useLoaderData();
  console.log(data);
  return <div>Users</div>;
};
```

```
export default Users;
```

- **Output:**